

# Reconstruction Metric - GPU Optimized

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```
# Configuración para visualización en PDF
import pandas as pd
import numpy as np
import sys
import os
from sae.tools.experiment_utils import get_metrics_dir

pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 100)
np.set_printoptions(linewidth=100, edgeitems=3)

os.environ['COLUMNS'] = '100'

print(" Configuración para PDF lista")
```

## Configuración para PDF lista

```
NAMING_META = {
    'variante': 'standard',          # Arquitectura del SAE
    'tecnica': 'l1-decay',          # L1 con penalty annealing
    'capa': 6,                      # Layer del modelo OthelloGPT
    'juegos': 500,                  # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt', # Path base para checkpoints
    'experiment_base_path': os.path.abspath('.././experiments'), # sae/experiments/
    'extras': {
        'expansion': 32,
        'sparsity_coef': 5e-4,
        'epochs': 200,
        'patience': 15
    }
}

print('\n Metadata de naming:')
for key, value in NAMING_META.items():
    print(f' {key}: {value}')
```

```

Metadata de naming:
  variante: standard
  tecnica: l1-decay
  capa: 6
  juegos: 1500
  base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
  experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
  extras: {'expansion': 32, 'sparsity_coef': 0.01, 'epochs': 1000, 'patience': 20, 'learning_rate': 0.001, 'sparsity_de

```

## 0.1 1. Setup e Imports

```

import numpy as np
import torch
import sys
from pathlib import Path
from tqdm import tqdm
import time

project_root = Path('../..').resolve()
sys.path.insert(0, str(project_root))

device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Device: {device}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
    print(f"Memoria: {torch.cuda.get_device_properties(0).total_memory / 1e9:.2f} GB")

```

```

Device: cuda
GPU: NVIDIA GeForce RTX 5060
Memoria: 8.55 GB

```

## 0.2 2. Cargar Datos

```

# Cargar activaciones
activations_path = project_root / "sae" / "activations" / "data" / "layer6_200games.npy"
activations = np.load(activations_path)

print(f"Activaciones del modelo:")
print(f"  Shape: {activations.shape}")
print(f"  Memoria: {activations.nbytes / (1024**2):.2f} MB")

```

```

Activaciones del modelo:
  Shape: (11800, 512)
  Memoria: 23.05 MB

```

```

# Cargar ground truth de BSPs
bsp_gt_path = project_root / "sae" / "metrics" / "02_data" / "bsp_ground_truth_200games.npy"
bsp_names_path = project_root / "sae" / "metrics" / "02_data" / "bsp_ground_truth_200games.names.npy"

bsp_ground_truth = np.load(bsp_gt_path)
bsp_names = np.load(bsp_names_path, allow_pickle=True)

print(f"\nGround truth BSPs:")
print(f"  Shape: {bsp_ground_truth.shape}")
print(f"  Total BSPs: {len(bsp_names)}")

```

Ground truth BSPs:  
Shape: (11800, 326)  
Total BSPs: 326

### 0.3 3. Cargar SAE y Extraer Features

```
from sae.models.sae import SparseAutoencoder
from sae.tools.naming_utils import get_checkpoint_dir, get_model_name, get_extras_id

input_dim = 512
hidden_dim = 16384

sae = SparseAutoencoder(input_dim, hidden_dim).to(device)
checkpoint_dir = get_checkpoint_dir(
    NAMING_META['base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos']
)
extras_id = get_extras_id(checkpoint_dir, NAMING_META['extras']) if NAMING_META.get('extras') else None

model_filename = get_model_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'best',
    extras_id=extras_id
)
model_path = checkpoint_dir / model_filename
checkpoint = torch.load(model_path, map_location=device, weights_only=False)
sae.load_state_dict(checkpoint['model_state_dict'])
sae.eval()

# Extraer info de epoch/val_mse según formato del checkpoint
metadata = checkpoint.get('metadata', {})
history = checkpoint.get('history', {})
epoch = checkpoint.get('epoch', history.get('best_epoch', metadata.get('estado', 'N/A')))
val_mse = checkpoint.get('val_mse', history.get('best_val_mse', 'N/A'))

print(f"SAE cargado:")
print(f"  Path: {model_path}")
print(f"  Input: {input_dim}, Hidden: {hidden_dim}")
print(f"  Expansion: {hidden_dim/input_dim}x")
print(f"  Epoch: {epoch}, Val MSE: {val_mse if val_mse == 'N/A' else f'{val_mse:.6f}'}")
print(f"  Extras ID: {extras_id}")
print(f"  Claves checkpoint: {list(checkpoint.keys())}")
```

SAE cargado:

Path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer\_06\models\sae\_standard\sae\_l1-decay\_standard\1500games\sae\_l1-decay\_standard\decay\_l6\_1500g\_ex1\_best.pt  
Input: 512, Hidden: 16384  
Expansion: 32.0x  
Epoch: best, Val MSE: N/A  
Extras ID: ex1  
Claves checkpoint: ['model\_state\_dict', 'config', 'metadata', 'history', 'optimizer\_state\_dict']

```
# Extraer features del SAE en GPU
print("Extrayendo features del SAE...")
activations_tensor = torch.from_numpy(activations).float().to(device)

with torch.no_grad():
    sae_features = torch.relu(sae.encoder(activations_tensor))

print(f"\nFeatures SAE:")
print(f"  Shape: {sae_features.shape}")
print(f"  Device: {sae_features.device}")
print(f"  Sparsity: {(sae_features == 0).float().mean():.2%}")
print(f"  Activaciones promedio: {(sae_features > 0).sum(dim=1).float().mean():.1f}")
```

Extrayendo features del SAE...

```
Features SAE:
  Shape: torch.Size([11800, 16384])
  Device: cuda:0
  Sparsity: 99.83%
  Activaciones promedio: 27.5
```

#### 0.4 4. Split Train/Test y Filtrar BSPs de Piezas

```
# Split: primeras 100 partidas = train, últimas 100 = test
n_moves = 59
split_idx = 100 * n_moves

sae_features_train = sae_features[:split_idx]
sae_features_test = sae_features[split_idx:]

bsp_gt_train = bsp_ground_truth[:split_idx]
bsp_gt_test = bsp_ground_truth[split_idx:]

print(f"Train set: {sae_features_train.shape}")
print(f"Test set: {sae_features_test.shape}")
```

```
Train set: torch.Size([5900, 16384])
Test set: torch.Size([5900, 16384])
```

```
# Filtrar BSPs de piezas (terminan en '1' o '2', no en '0')
bsp_pieces_indices = []
bsp_pieces_names = []

for i, name in enumerate(bsp_names):
    if len(name) == 6 and name.startswith('BSP') and not name.endswith('0'):
        bsp_pieces_indices.append(i)
        bsp_pieces_names.append(name)

bsp_pieces_indices = np.array(bsp_pieces_indices)

# Aplicar filtro y convertir a tensores GPU
bsp_gt_train_pieces = torch.from_numpy(bsp_gt_train[:, bsp_pieces_indices]).bool().to(device)
bsp_gt_test_pieces = torch.from_numpy(bsp_gt_test[:, bsp_pieces_indices]).bool().to(device)

print(f"\nBSPs para Reconstruction:")
print(f"  Total BSPs de piezas: {len(bsp_pieces_indices)}")
print(f"  Train shape: {bsp_gt_train_pieces.shape}")
print(f"  Test shape: {bsp_gt_test_pieces.shape}")
```

BSPs para Reconstruction:

Total BSPs de piezas: 256  
Train shape: torch.Size([5900, 256])  
Test shape: torch.Size([5900, 256])

## 0.5 5. Funciones GPU-Optimizadas

```
def fast_precision_score_gpu(y_true, y_pred, eps=1e-8):
    """
    Precisión vectorizada en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions, n_candidates) booleano

    Returns:
        precision_scores: Tensor (n_candidates,)
    """
    y_true_expanded = y_true.unsqueeze(1)

    tp = (y_true_expanded & y_pred).sum(dim=0).float()
    fp = (~y_true_expanded & y_pred).sum(dim=0).float()

    precision = tp / (tp + fp + eps)

    return precision

def fast_f1_score_gpu(y_true, y_pred, eps=1e-8):
    """
    F1 score vectorizado en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions,) booleano

    Returns:
        f1: Float
    """
    tp = (y_true & y_pred).sum().float()
    fp = (~y_true & y_pred).sum().float()
    fn = (y_true & ~y_pred).sum().float()

    precision = tp / (tp + fp + eps)
    recall = tp / (tp + fn + eps)

    f1 = 2 * (precision * recall) / (precision + recall + eps)

    return f1.item()

print(" Funciones GPU definidas")
```

Funciones GPU definidas

## 0.6 6. Fase 1: Identificar Features de Alta Precisión (GPU)

Para cada BSP, encontrar features con precisión 0.95 en el train set.

```

def identify_high_precision_features_gpu(
    sae_features_train,
    bsp_gt_train,
    f_max,
    precision_threshold=0.95,
    significance_threshold=10,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    feature_batch_size=512,
    verbose=True
):
    """
    Para cada (threshold t, feature f, BSP b): determina si f es clasificador
    de alta precisión para b al threshold t.

    Criterios (igual que implementación oficial):
    - precision(f, b, t) >= precision_threshold (default 0.95)
    - on_count(f, t) >= significance_threshold (default 10)

    Args:
        sae_features_train: Tensor (P, n_features) GPU
        bsp_gt_train: Tensor (P, n_bsps) GPU, bool
        f_max: Tensor (n_features,) GPU - calculado FUERA para
            garantizar identidad exacta con Fase 2
        precision_threshold: float, default 0.95
        significance_threshold: int, mín. activaciones requeridas (default 10)
        thresholds: lista de fracciones t [0, 1)
        feature_batch_size: int, features por mini-batch interno

    Returns:
        hp_mask_TFB: BoolTensor (T, n_active, n_bsps)
        active_feature_indices: LongTensor (n_active,)
    """
    n_positions, n_features = sae_features_train.shape
    n_bsps = bsp_gt_train.shape[1]
    device = sae_features_train.device

    thresholds_T = torch.tensor(thresholds, device=device)
    n_thresholds = len(thresholds_T)

    # Features vivas (f_max > 0)
    active_feature_indices = (f_max > 0).nonzero(as_tuple=True)[0]
    n_active = len(active_feature_indices)

    if verbose:
        print("=" * 60)
        print("FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN")
        print("=" * 60)
        print(f"Precision threshold: {precision_threshold}")
        print(f"Significance threshold: >= {significance_threshold} activaciones")
        print(f"Features activas: {n_active}/{n_features} ({n_active/n_features*100:.1f}%)")
        print(f"BSPs: {n_bsps}")
        print(f"Thresholds: {thresholds}")
        print("=" * 60)

    # Thresholds absolutos: thresh_TF[t, f] = t * f_max[f]
    active_f_max = f_max[active_feature_indices] # (n_active,)
    thresholds_TF = thresholds_T[:, None] * active_f_max # (T, n_active)

    # Acumuladores sobre todo el train set
    on_count_TF = torch.zeros(n_thresholds, n_active, device=device)
    tp_count_TFB = torch.zeros(n_thresholds, n_active, n_bsps, device=device)

```

```

bsp_float_PB = bsp_gt_train.float() # (P, B)

n_iters = (n_active + feature_batch_size - 1) // feature_batch_size

for fi in tqdm(range(n_iters), desc="Fase 1"):
    f_start = fi * feature_batch_size
    f_end = min(f_start + feature_batch_size, n_active)

    global_idx = active_feature_indices[f_start:f_end]
    acts_PF = sae_features_train[:, global_idx] # (P, bs)
    thresh_TF = thresholds_TF[:, f_start:f_end] # (T, bs)

    # active_TFP[t, f, p] = True si acts[p,f] > thresh[t,f]
    active_TFP = acts_PF.T.unsqueeze(0) > thresh_TF.unsqueeze(2) # (T, bs, P)

    # Conteo de activaciones por (threshold, feature)
    on_count_TF[:, f_start:f_end] += active_TFP.sum(dim=2).float()

    # True positives: f activa Y BSP verdadera
    # tp[t, f, b] = Σ_p active[t,f,p] * bsp[p,b]
    tp_count_TFB[:, f_start:f_end, :] += torch.einsum(
        'tfp,pb->tfb', active_TFP.float(), bsp_float_PB
    )

# Precisión: tp / on_count
eps = 1e-8
precision_TFB = tp_count_TFB / (on_count_TF.unsqueeze(2) + eps) # (T, F, B)

# Máscara: alta precisión AND significancia suficiente
sig_mask_TF1 = (on_count_TF >= significance_threshold).unsqueeze(2) # (T, F, 1)
hp_mask_TFB = (precision_TFB >= precision_threshold) & sig_mask_TF1 # (T, F, B)

if verbose:
    hp_per_t = hp_mask_TFB.sum(dim=(1, 2)) # (T,)
    hp_per_b = hp_mask_TFB.sum(dim=1) # (T, B)
    best_t = hp_per_t.argmax().item()
    print(f"\nPares (feature, BSP) de alta precisión por threshold:")
    for i, t in enumerate(thresholds):
        marker = " ← más pares" if i == best_t else ""
        print(f" t={t:.1f}: {hp_per_t[i].item():5d} pares{marker}")
    print(f"\nFeatures HP por BSP en t={thresholds[best_t]:.1f}:")
    print(f" Media: {hp_per_b[best_t].float().mean():.1f}")
    print(f" Máx: {hp_per_b[best_t].max().item()}")
    print(f" BSPs sin features: {(hp_per_b[best_t] == 0).sum().item()}")

    return hp_mask_TFB, active_feature_indices

print(" Fase 1 rediseñada: umbral de significancia (10) + estructura (T, F, B)")

```

Fase 1 rediseñada: umbral de significancia (10) + estructura (T, F, B)

```

THRESHOLDS = [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]

# f_max calculado UNA SOLA VEZ aquí - se pasa a Fase 1 y Fase 2
f_max = sae_features_train.max(dim=0)[0] # (n_features,)
print(f"f_max calculado: {(f_max > 0).sum().item()} features activas de {f_max.shape[0]}")

start_time = time.time()

hp_mask_TFB, active_feature_indices = identify_high_precision_features_gpu(
    sae_features_train,

```

```

    bsp_gt_train_pieces,
    f_max,
    precision_threshold=0.95,
    significance_threshold=10,
    thresholds=THRESHOLDS,
    feature_batch_size=512,
    verbose=True
)

phase1_time = time.time() - start_time

print(f"\nTiempo Fase 1: {phase1_time:.2f} seg")
print(f"hp_mask_TFB shape: {hp_mask_TFB.shape} (T, n_active, n_bsps)")

```

```

f_max calculado: 11494 features activas de 16384
=====
FASE 1: IDENTIFICAR FEATURES DE ALTA PRECISIÓN
=====
Precision threshold:      0.95
Significance threshold:   >= 10 activaciones
Features activas:        11494/16384 (70.2%)
BSPs:                    256
Thresholds:              [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]
=====

```

```
Fase 1: 100%|      | 23/23 [00:00<00:00, 92.08it/s]
```

Pares (feature, BSP) de alta precisión por threshold:

```

t=0.0:    2 pares
t=0.1:    0 pares
t=0.2:    6 pares
t=0.3:   23 pares
t=0.4:   33 pares
t=0.5:   68 pares
t=0.6:   91 pares
t=0.7:  107 pares
t=0.8:  124 pares ← más pares
t=0.9:  117 pares

```

Features HP por BSP en t=0.8:

```

Media:      0.5
Máx:       11
BSPs sin features: 218

```

Tiempo Fase 1: 0.32 seg

```
hp_mask_TFB shape: torch.Size([10, 11494, 256]) (T, n_active, n_bsps)
```

## 0.7 7. Fase 2: Reconstruir Tableros en Test Set (GPU)

```

def reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test,
    hp_mask_TFB,
    active_feature_indices,
    f_max,
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],

```



```

verbose=True
):
    """
    Fase 2: barrido simultáneo de thresholds T → toma el mejor F1.

    Para cada threshold t:
    1. Binarizar features de test: active[p, f] = acts[p, f] > t * f_max[f]
    2. Predicción por (posición, BSP):
        predict[p, b] = OR_f ( active[p,f] AND hp_mask[t, f, b] )
        implementado como: (active_PF @ hp_FB) > 0
    3. F1 global micro-averaged (acumula TP/FP/FN sobre todas las posiciones y BSPs)

    Retorna el max F1 sobre todos los thresholds (Eq. 5 del paper: max_t).

    Args:
        sae_features_test:      Tensor (P, n_features) GPU
        bsp_gt_test:           Tensor (P, n_bsps) GPU, bool
        hp_mask_TFB:           BoolTensor (T, n_active, n_bsps) de Fase 1
        active_feature_indices: LongTensor (n_active,) de Fase 1
        f_max:                  Tensor (n_features,) GPU - MISMO objeto que Fase 1
        thresholds:             lista de fracciones (debe coincidir con Fase 1)

    Returns:
        best_f1:                float, max F1 sobre thresholds
        best_threshold:         float, threshold óptimo
        f1_per_threshold:       np.array (T,), F1 por threshold
    """
    device = sae_features_test.device
    n_thresholds = len(thresholds)
    thresholds_T = torch.tensor(thresholds, device=device)

    # Activaciones del test solo para features activas
    test_active_PF = sae_features_test[:, active_feature_indices] # (P, n_active)
    active_f_max = f_max[active_feature_indices] # (n_active,)

    gt_PB = bsp_gt_test.bool() # (P, B)
    eps = 1e-8

    f1_scores = torch.zeros(n_thresholds, device=device)
    precision_T = torch.zeros(n_thresholds, device=device)
    recall_T = torch.zeros(n_thresholds, device=device)

    if verbose:
        print("=" * 60)
        print("FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS")
        print("=" * 60)
        print(f"{'t':>5} {'F1':>8} {'P':>8} {'R':>8} {'TP':>8} {'FP':>8} {'FN':>8}")
        print("-" * 60)

    for t_idx in range(n_thresholds):
        t = thresholds[t_idx]

        # Threshold absoluto por feature: t * f_max[f]
        abs_thresh_F = thresholds_T[t_idx] * active_f_max # (n_active,)

        # Binarizar features en test
        active_PF = test_active_PF > abs_thresh_F.unsqueeze(0) # (P, n_active)

        # predict[p, b] = any( active[p,f] AND hp[t,f,b] )
        # = (active_PF.float() @ hp_mask[t].float()) > 0
        hp_FB = hp_mask_TFB[t_idx].float() # (n_active, B)
        predictions_PB = (active_PF.float() @ hp_FB) > 0 # (P, B)

```

```

# Métricas globales micro-averaged
tp = (predictions_PB & gt_PB).sum().float()
fp = (predictions_PB & ~gt_PB).sum().float()
fn = (~predictions_PB & gt_PB).sum().float()

p = tp / (tp + fp + eps)
r = tp / (tp + fn + eps)
f1 = 2 * p * r / (p + r + eps)

f1_scores[t_idx] = f1
precision_T[t_idx] = p
recall_T[t_idx] = r

if verbose:
    print(f"{t:>5.1f} {f1.item():>8.4f} {p.item():>8.4f} {r.item():>8.4f}"
          f" {tp.item():>8.0f} {fp.item():>8.0f} {fn.item():>8.0f}")

best_idx = f1_scores.argmax().item()
best_f1 = f1_scores[best_idx].item()
best_threshold = thresholds[best_idx]

if verbose:
    print("=" * 60)
    print(f"Mejor threshold: t={best_threshold:.1f}")
    print(f"Reconstruction Score (max_t F1): {best_f1:.4f}")
    print("=" * 60)

return best_f1, best_threshold, f1_scores.cpu().numpy()

print(" Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max_t")

```

Fase 2 rediseñada: barrido T simultáneo + F1 global micro-averaged + max\_t

```

start_time = time.time()

reconstruction_score, best_threshold, f1_per_threshold = reconstruct_boards_gpu(
    sae_features_test,
    bsp_gt_test_pieces,
    hp_mask_TFB,
    active_feature_indices,
    f_max, # mismo tensor calculado antes de Fase 1
    thresholds=THRESHOLDS,
    verbose=True
)

phase2_time = time.time() - start_time
total_time = phase1_time + phase2_time

print(f"\nTiempo Fase 2: {phase2_time:.2f} seg")
print(f"Tiempo total: {total_time:.2f} seg ({total_time/60:.2f} min)")

```

```

=====
FASE 2: RECONSTRUCCIÓN - BARRIDO DE THRESHOLDS
=====

```

| t   | F1     | P      | R      | TP  | FP | FN     |
|-----|--------|--------|--------|-----|----|--------|
| 0.0 | 0.0000 | 1.0000 | 0.0000 | 2   | 0  | 401198 |
| 0.1 | 0.0000 | 0.0000 | 0.0000 | 0   | 0  | 401200 |
| 0.2 | 0.0022 | 0.9449 | 0.0011 | 446 | 26 | 400754 |
| 0.3 | 0.0040 | 0.9795 | 0.0020 | 814 | 17 | 400386 |
| 0.4 | 0.0039 | 0.9739 | 0.0020 | 783 | 21 | 400417 |

|     |        |        |        |      |    |        |
|-----|--------|--------|--------|------|----|--------|
| 0.5 | 0.0062 | 0.9665 | 0.0031 | 1242 | 43 | 399958 |
| 0.6 | 0.0085 | 0.9695 | 0.0043 | 1715 | 54 | 399485 |
| 0.7 | 0.0091 | 0.9730 | 0.0046 | 1840 | 51 | 399360 |
| 0.8 | 0.0091 | 0.9903 | 0.0046 | 1832 | 18 | 399368 |
| 0.9 | 0.0076 | 1.0000 | 0.0038 | 1532 | 0  | 399668 |

```
=====
Mejor threshold: t=0.7
Reconstruction Score (max_t F1): 0.0091
=====
```

```
Tiempo Fase 2: 0.17 seg
Tiempo total: 0.49 seg (0.01 min)
```

## 0.8 8. Análisis de Resultados

```
import matplotlib.pyplot as plt

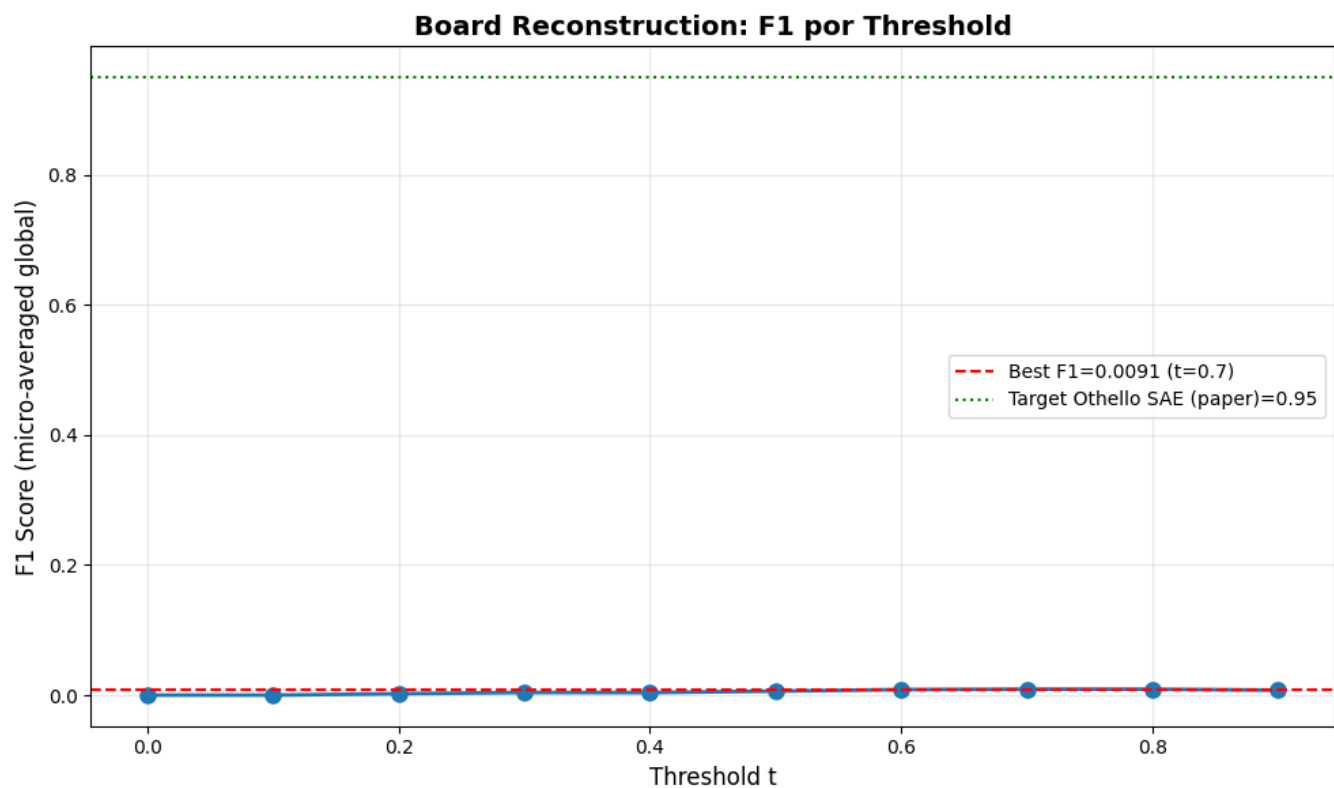
print("=" * 60)
print("BARRIDO DE THRESHOLDS - F1 POR THRESHOLD")
print("=" * 60)
for i, (t, f1) in enumerate(zip(THRESHOLDS, f1_per_threshold)):
    marker = " ← MEJOR" if i == f1_per_threshold.argmax() else ""
    print(f"t={t:.1f}: F1={f1:.4f}{marker}")
print("=" * 60)

plt.figure(figsize=(10, 6))
plt.plot(THRESHOLDS, f1_per_threshold, marker='o', linewidth=2, markersize=8)
plt.axhline(y=reconstruction_score, color='r', linestyle='--',
            label=f'Best F1={reconstruction_score:.4f} (t={best_threshold:.1f})')
plt.axhline(y=0.95, color='g', linestyle=':', label='Target Othello SAE (paper)=0.95')
plt.xlabel('Threshold t', fontsize=12)
plt.ylabel('F1 Score (micro-averaged global)', fontsize=12)
plt.title('Board Reconstruction: F1 por Threshold', fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()

print(f"\n Threshold óptimo: t={best_threshold:.1f}")
print(f" Reconstruction Score: {reconstruction_score:.4f}")
```

```
=====
BARRIDO DE THRESHOLDS - F1 POR THRESHOLD
=====
```

```
t=0.0: F1=0.0000
t=0.1: F1=0.0000
t=0.2: F1=0.0022
t=0.3: F1=0.0040
t=0.4: F1=0.0039
t=0.5: F1=0.0062
t=0.6: F1=0.0085
t=0.7: F1=0.0091 ← MEJOR
t=0.8: F1=0.0091
t=0.9: F1=0.0076
=====
```



Threshold óptimo: t=0.7  
Reconstruction Score: 0.0091

```
# Estadísticas sobre features de alta precisión por BSP
hp_per_bsp = hp_mask_TFB.sum(dim=1) # (T, n_bsps)
best_t_idx = torch.tensor(f1_per_threshold).argmax().item()
n_features_per_bsp = hp_per_bsp[best_t_idx].cpu().numpy()

print("="*60)
print("ESTADÍSTICAS - FEATURES POR BSP")
print("="*60)
print(f"Total BSPs: {len(n_features_per_bsp)}")
print(f"Features promedio por BSP: {np.mean(n_features_per_bsp):.1f}")
print(f"Features mediana por BSP: {np.median(n_features_per_bsp):.1f}")
print(f"BSPs con 0 features: {np.sum(n_features_per_bsp == 0)}")
print(f"BSPs con 1+ features: {np.sum(n_features_per_bsp > 0)}")
print(f"BSPs con 5+ features: {np.sum(n_features_per_bsp >= 5)}")
print(f"BSPs con 10+ features: {np.sum(n_features_per_bsp >= 10)}")
print(f"Max features en una BSP: {np.max(n_features_per_bsp)}")
print("="*60)
```

```
=====
ESTADÍSTICAS - FEATURES POR BSP
=====
Total BSPs: 256
Features promedio por BSP: 0.4
Features mediana por BSP: 0.0
BSPs con 0 features: 221
BSPs con 1+ features: 35
BSPs con 5+ features: 7
BSPs con 10+ features: 1
Max features en una BSP: 11
=====
```

```
# Comparación con resultados del paper
print("="*60)
print("COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)")
print("="*60)
print(f"{'Modelo':<20} {'Reconstruction':<15}")
print("-"*60)
print(f"{'SAE random':<20} {'0.08':<15}")
print(f"{'SAE trained (paper)':<20} {'0.95':<15} ← Objetivo")
print(f"{'Linear probe':<20} {'0.99':<15}")
print(f"{'Nuestro SAE':<20} {'reconstruction_score:<15.4f'} ← Resultado")
print("="*60)

if reconstruction_score >= 0.90:
    print("\n Excelente: Muy cercano al objetivo del paper")
elif reconstruction_score >= 0.80:
    print("\n Bueno: Por encima de random pero debajo del objetivo")
else:
    print("\n Bajo: Revisar implementación o entrenamiento del SAE")
```

```
=====
COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)
=====
Modelo                Reconstruction
-----
SAE random            0.08
SAE trained (paper)   0.95                ← Objetivo
Linear probe          0.99
Nuestro SAE           0.0091            ← Resultado
=====
```

Bajo: Revisar implementación o entrenamiento del SAE

## 0.9 9. Guardar Resultados

```
results = {
    'reconstruction_score': reconstruction_score,
    'best_threshold': best_threshold,
    'f1_per_threshold': f1_per_threshold,
    'thresholds': np.array(THRESHOLDS),
    'phase1_time_seconds': phase1_time,
    'phase2_time_seconds': phase2_time,
    'total_time_seconds': total_time,
    'bsp_names': bsp_pieces_names,
}

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)
output_path = metrics_dir / "reconstruction_results.npz"
np.savez(output_path, **results)

print(f" Resultados guardados en: {output_path}")
print(f"\nContenido:")
print(f" reconstruction_score: {reconstruction_score:.4f}")
print(f" best_threshold:      {best_threshold:.1f}")
```

```

print(f"  fl_per_threshold:      {fl_per_threshold}")

# Comparación con el paper
print()
print("=" * 60)
print("COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)")
print("=" * 60)
print(f"{'Modelo':<25} {'Reconstruction':>14}")
print("-" * 40)
print(f"{'SAE random':<25} {'0.08':>14}")
print(f"{'SAE trained (paper)':<25} {'0.95':>14} ← Objetivo Othello")
print(f"{'Linear probe':<25} {'0.99':>14}")
print(f"{'Nuestro SAE':<25} {reconstruction_score:>14.4f} ← Resultado")
print("=" * 60)

if reconstruction_score >= 0.90:
    print("\n Excelente: Muy cercano al objetivo del paper")
elif reconstruction_score >= 0.50:
    print("\n~ Moderado: Por encima de random, revisar SAE")
else:
    print("\n Bajo: Revisar entrenamiento del SAE o calidad de las activaciones")

```

Resultados guardados en: c:\Users\Esposa\Documents\Repos\othello\_world\sae\experiments\layer\_06\sae\_standard\sae\_l1-decay\_standard\1500games\ex1\metrics\reconstruction\_results.npz

Contenido:

```

reconstruction_score: 0.0091
best_threshold:      0.7
fl_per_threshold:    [9.9700401e-06 0.0000000e+00 2.2207173e-03 4.0494390e-03 3.8954834e-03 6.1716586e-03 8.5118217e-03 9.1294516e-03 9.0906834e-03 7.6080374e-03]

```

```

=====
COMPARACIÓN CON PAPER (Karvonen et al., NeurIPS 2024)
=====
Modelo                Reconstruction
-----
SAE random            0.08
SAE trained (paper)   0.95 ← Objetivo Othello
Linear probe          0.99
Nuestro SAE           0.0091 ← Resultado
=====

```

Bajo: Revisar entrenamiento del SAE o calidad de las activaciones

```

import subprocess
from sae.tools.experiment_utils import get_report_name

output_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],

```

```

    'reconstruction',
    extras_id=extras_id
)

notebook_name = 'reconstruction_gpu_optimized_v2.ipynb'
output_path = output_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

result = subprocess.run(
    f'quarto render {notebook_name} --to pdf --output-dir "{output_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)

```

Generando PDF con Quarto...

Archivo de salida: c:\Users\Esposa\Documents\Repos\othello\_world\sae\experiments\layer\_06\sae\_standard\sae\_l1-decay\_standard\1500games\ex1\metrics\sae\_standard\_l1-decay\_l6\_1500g\_ex1\_reconstruction.pdf

PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello\_world\sae\experiments\layer\_06\sae\_standard\sae\_l1-decay\_standard\1500games\ex1\metrics\sae\_standard\_l1-decay\_l6\_1500g\_ex1\_reconstruction.pdf